

Sub-parsec binary SMBHs imaged in NGC 5044

VLBA resolves two radio cores, confirming a pc-scale binary in a group-central galaxy.

arXiv:2609.22433; VLBA 4.9–23.6 GHz; NGC 5044 group.

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Background

Binary supermassive black holes are expected during galaxy mergers, but direct imaging at sub-parsec scales remains rare. NGC 5044 is the central galaxy of a nearby group, offering a unique laboratory.

Only one pc-scale binary SMBH was previously known in a cluster-central elliptical; more are needed to test merger scenarios.

What this paper is selling

The paper reports VLBA imaging that resolves two compact radio cores separated by 0.4 pc in NGC 5044. Their flat-to-inverted spectra confirm both as active SMBHs, establishing a second pc-scale binary in a group-central galaxy.

Key results to remember

- Two compact radio cores, C1 and C2, separated by 0.4 pc, are resolved.
- Both cores show flat-to-inverted spectra, typical of AGN cores.
- This is the second pc-scale binary SMBH in a cluster-central elliptical.

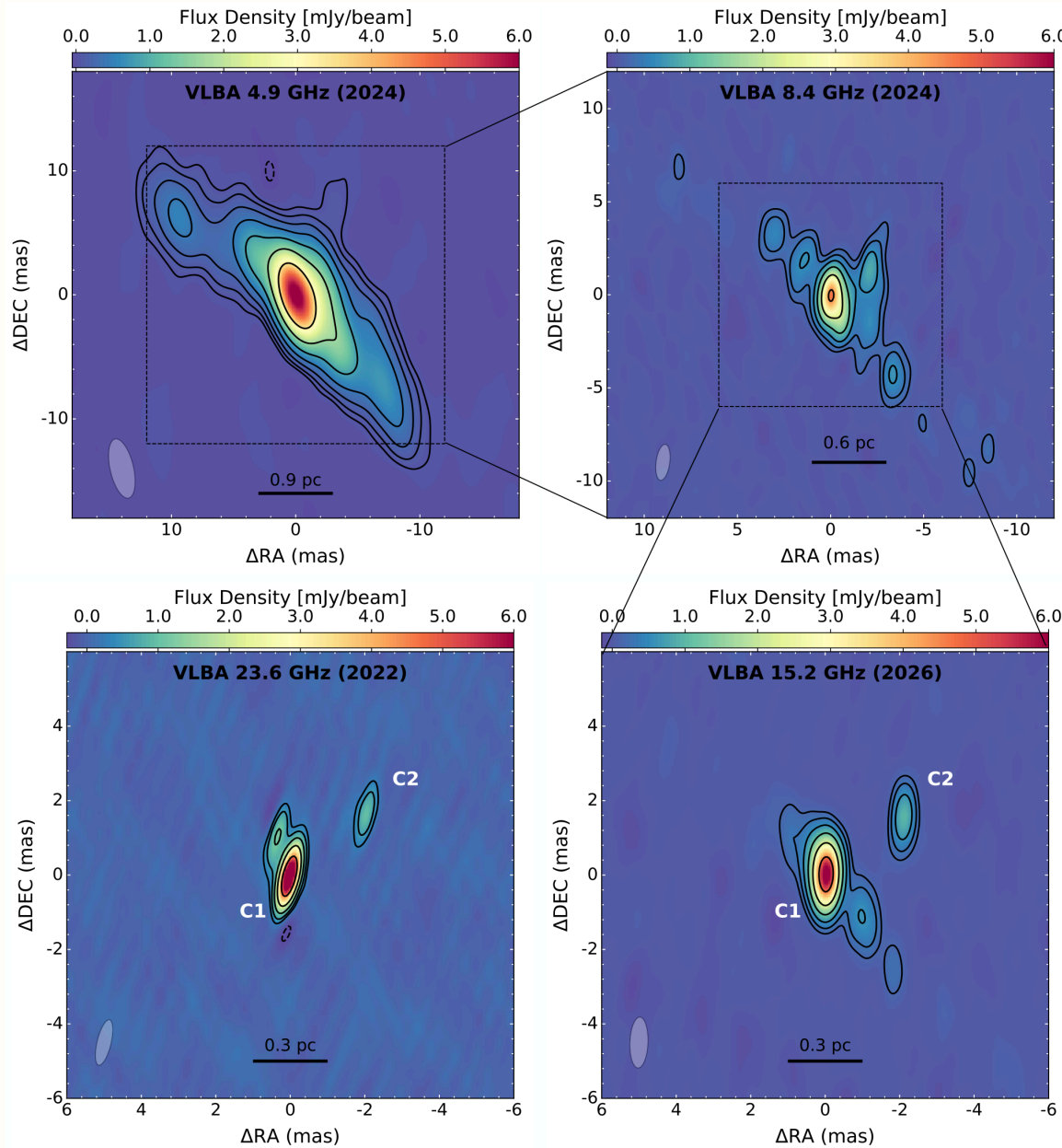


Figure 1 | VLBA images at increasing frequencies resolve the core into two components, C1 and C2, separated by 0.4 pc.

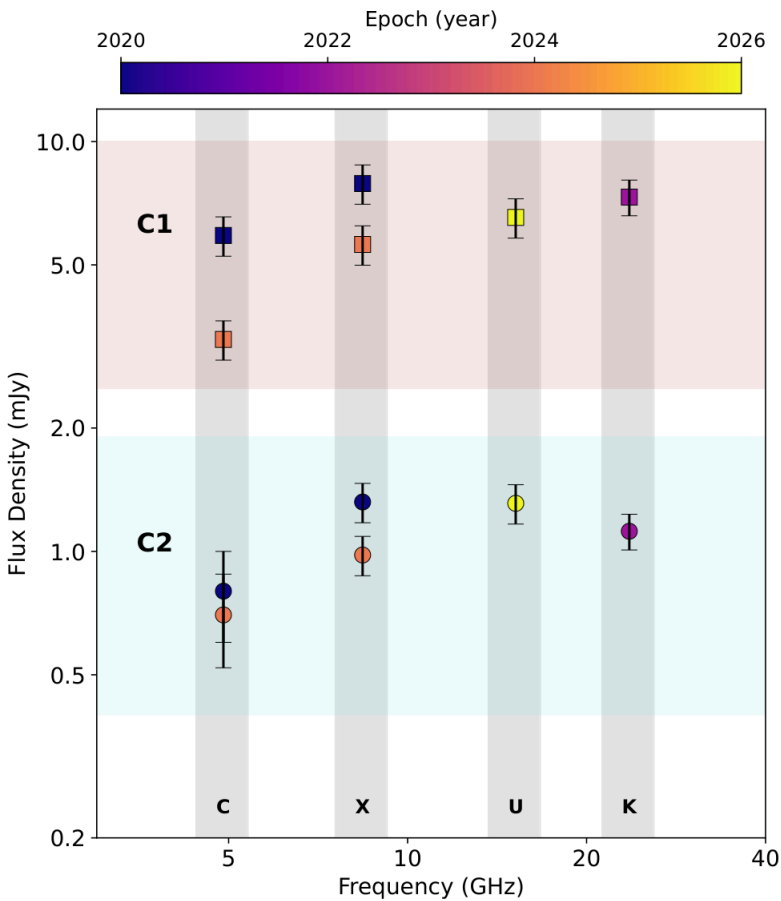


Figure 2 | Radio spectra show C1 and C2 have flat-to-inverted spectra, while the jet knot has a steep spectrum.